

Q1.A real estate company collected the following training data:(15Marks -20 Mins)

House No Area (sq.ft) Bedrooms Price (₹ Lakhs)

1 1000 2 50

2 1500 3 75

3 800 2 40

4 1200 3 60

5 2000 4 90

(i) Construct a linear regression model of the form

(ii) Estimate the regression coefficients.

(iii) Interpret the meaning of each coefficient.

```
main1.py x
1 from sklearn.linear_model import LinearRegression
2 X = [
3     [1000, 2],
4     [1500, 3],
5     [800, 2],
6     [1200, 3],
7     [2000, 4]
8 ]
9
10 y = [50, 75, 40, 60, 90]
11 model = LinearRegression()
12 model.fit(X, y)
13 print("Intercept:", model.intercept_)
14 print("Area Coefficient:", model.coef_[0])
15 print("Bedroom Coefficient:", model.coef_[1])
16 print("\nRegression Equation:")
17 print(f"Price = {model.intercept_:.2f} + {model.coef_[0]:.4f}*Area + {model.coef_[1]:.2f}*Bedrooms")
```

Output:

```
Intercept: 8.50467289719628
Area Coefficient: 0.0425233644859813
Bedroom Coefficient: -0.2803738317757003

Regression Equation:
Price = 8.50 + 0.0425*Area + -0.28*Bedrooms

Process finished with exit code 0
```

Q2.An email system uses the following feature values:(15Marks -20 Mins)

Email Word Frequency (Offer) Word Frequency (Win) Spam (1/0)

1 3 2 1

2 2 1 1

3 1 0 0

4 3 3 1

5 0 1 0

(i) Construct a linear classification model.

(ii) Determine the decision boundary.

(iii) Classify a new email with: Offer = 2, Win = 1.

```
main1.py ×
1 from sklearn.linear_model import LogisticRegression
2 X = [
3     [3, 2], [2, 1],
4     [1, 0],
5     [3, 3],
6     [0, 1]
7 ]
8 y = [1, 1, 0, 1, 0]
9 model = LogisticRegression()
10 model.fit(X, y)
11 print("Intercept:", model.intercept_[0])
12 print("Offer Coefficient:", model.coef_[0][0])
13 print("Win Coefficient:", model.coef_[0][1])
14 new_email = [[2, 1]]
15 prediction = model.predict(new_email)
16 print("\nPrediction:")
17 if prediction[0] == 1:
18     print("Spam")
19 else:
20     print("Not Spam")
```

Output:

```
C:\Users\prave\PycharmProjects1\explat1\.venv\Scripts\python.exe C:\Users\prave\PycharmProjects1\explat1\main1.py
Intercept: -1.7757916317080757
Offer Coefficient: 0.9278135145654106
Win Coefficient: 0.5431298300308649

Prediction:
Spam

Process finished with exit code 0
```

Q3.A medical dataset is given:(15Marks -20 Mins)

Patient Fever (1/0) Headache (1/0) Disease (Yes/No)

1 1 1 Yes

2 1 0 Yes

3 0 1 No

4 1 1 Yes

5 0 0 No

(i) Compute prior probabilities.

(ii) Compute conditional probabilities.

(iii) Estimate the probability that a patient with Fever = 1 and Headache = 0 has the disease.

```
main1.py x
1 P_yes = 3 / 5
2 P_no = 2 / 5
3 P_fever_yes = 3 / 3
4 P_headache_yes = 1 / 3
5
6 P_fever_no = 0 / 2
7 P_headache_no = 1 / 2
8
9 yes = P_yes * P_fever_yes * P_headache_yes
10 no = P_no * P_fever_no * P_headache_no
11
12 print("Probability of Yes:", yes)
13 print("Probability of No:", no)
14
15 if yes > no:
16     print("Prediction: Disease = Yes")
17 else:
18     print("Prediction: Disease = No")
```

Output:

```
C:\Users\prave\PycharmProjects1\explat1\.venv\Scripts\python.exe C:\Users\prave\PycharmProjects1\explat1\main1.py
Probability of Yes: 0.19999999999999998
Probability of No: 0.0
Prediction: Disease = Yes

Process finished with exit code 0
```